

9/10 8/26 8-30



UNIVERSITY OF ILLINOIS PRESS, URBANA, 1964

# PRIDE AND POWER

*The Rationale of the Space Program*

*by Vernon Van Dyke*

VERNON VAN DYKE, PRIDE AND POWER;  
THE RATIONALE OF THE SPACE  
PROGRAM (1964)

INTRODUCTION

1

Though the space program is widely regarded as scientific and technological, it is also obviously political. It is implemented by government, partly for political reasons, and it involves both international competition and domestic controversy.

The controversy, at its core, is over the question what the reasons for the space program are and whether they justify the costs. Manned space flight is the chief issue, particularly the effort to get a man to the moon and back in this decade. How important is it that the trip occur and that it occur on so urgent a basis? How important is it that we beat the Russians or at least avoid being beaten? Apart from this issue, what other reasons for the space program operate, and how cogent are they? There is less controversy over governmental arrangements and activities for implementing the program, but enough to be significant nevertheless.

The fact of controversy over the rationale of the space program indicates a major purpose of the chapters that follow: to contribute to thought on the question why we have the program and on the basis of what principles it should be managed.

Other purposes can be served simultaneously. How decisions get made and to what extent the process is or can be rational are questions of universal application in political science. Decisions are central to political behavior—decisions by individual voters, by administrators or office holders, by governmental agencies of all sorts—and they can be studied in many different ways. In what

follows I propose to study in a macroscopic way the major decisions that have been made relating to the American space program. I say macroscopic because I will not make a meticulous effort to determine precisely how much influence specific individuals have had on decisions and precisely what their personal thought processes have been; rather, I propose to engage in a more general analysis of the reasons given to explain or justify the decisions that have been made and to identify the categories of persons or groups endorsing or responding to the various reasons. I will pay special attention to attitudes in the Executive branch of government and in Congress. I do not expect to arrive at a definitive answer to the question whether major governmental decisions can be rational, or even whether decisions concerning space have been rational. But I hope to contribute to consideration of the problem. I have in mind Charles E. Lindblom's analysis of "The Science of 'Muddling Through,'" and will refer to it again in the concluding chapter.

Still another purpose relates to the general problem that can figuratively be described as the failure of the United States to know its own mind. Especially in connection with foreign policy, virtually every crisis that the country confronted from the time of the Spanish-American War to World War II found the President and his advisers without a well-thought-out conception of the purposes (the goals, values, interests, motives<sup>1</sup>) that ought to be pursued. Time and again, therefore, improvisation has occurred, and goals have been pursued that had little or nothing to do with those previously proclaimed. Each crisis, each surprise, has called up a new, or at least a different, conception of goals, and each new or different conception has later been superseded by another. The characteristic is illustrated by the space program. Sputnik I and its successors led the United States to adopt goals (to act for reasons) some of which had rarely been mentioned theretofore. In particular, the idea that we should act so as to maintain or enhance our national pride and that this requires us to beat the Russians in scientific and technological achievement is a new and different motive called up by Sputnik; and that it may go the way of former major objectives is suggested by the late President's proposal of September, 1963, that instead of racing the Russians to the moon we should join with them in a cooperative program.

<sup>1</sup> Vernon Van Dyke, "Values and Interests," *American Political Science Review*, 56 (September, 1962), 567-76.

I will suggest an explanation of the apparent inconstancy of American purposes in Chapter 11.

### Reasons Commonly Cited for the Space Program

Though the reasons for the space program will be examined in detail in later chapters, it may be helpful at the outset to provide a sample of them. One of the early efforts to identify them coherently was made in 1958 in an "Introduction to Outer Space," prepared by President Eisenhower's Science Advisory Committee, under the chairmanship of Dr. James R. Killian, Jr. It describes four factors that give "importance, urgency, and inevitability to the advancement of space technology."

The first of these factors is the compelling urge of man to explore and to discover, the thrust of curiosity that leads men to try to go where no one has gone before. Most of the surface of the earth has now been explored and men now turn to the exploration of outer space as their next objective.

Second, there is the defense objective for the development of space technology. We wish to be sure that space is not used to endanger our security. If space is to be used for military purposes, we must be prepared to use space to defend ourselves.

Third, there is the factor of national prestige. To be strong and bold in space technology will enhance the prestige of the United States among the peoples of the world and create added confidence in our scientific, technological, industrial, and military strength.

Fourth, space technology affords new opportunities for scientific observation and experiment which will add to our knowledge and understanding of the earth, the solar system, and the universe.

In recommending the legislation that created NASA (the National Aeronautics and Space Administration), President Eisenhower listed the purposes of the organization. Among other things it was to contribute to "the preservation of the role of the United States as a leader in aeronautical and space science and technology." And this purpose is among those endorsed by Congress in the National Aeronautics and Space Act itself. Prior to taking office, President-elect Kennedy appointed an Ad Hoc Committee on Space, with Dr. Jerome B. Wiesner as Chairman, and its report included statements concerning the goals of a space program.

In addition to the need to develop ballistic missiles to provide for our military security, there are five principal motivations for desiring a vital, effective space program. . . .

First, there is the factor of national prestige. . . . During the next few years, the prestige of the United States will in part be determined by the leadership we demonstrate in space activities. . . .

Second, we believe that some space developments, in addition to missiles, can contribute much to our national security. . . .

Third, the development of space vehicles affords new opportunities for scientific observation and experiment—adding to our knowledge and understanding of the earth, the solar system, and the universe. . . .

Fourth, there are a number of important practical non-military applications of space technology—among them satellite communications and broadcasting, satellite navigation and geodesy, meteorological reconnaissance, and satellite mapping—which can make important contributions to our civilian efforts and to our economy.

Finally, space activities . . . offer exciting possibilities for international cooperation with all the nations of the world. The very ambitious and long-range space projects would prosper if they could be carried out in an atmosphere of cooperation as projects of all mankind instead of in the present atmosphere of national competition.

James E. Webb, Administrator of NASA, has made a number of statements of the rationale of the program that he directs. One of them is as follows, and others will be noted in subsequent chapters. "That is our objective—to develop superior competence in space which will be available for any national purpose which may be required, whether it be the peaceful utilization of space for the benefit of mankind, or to keep the peace and forestall its exploitation by any nation for aggressive purposes. . . . We will . . . occupy a position second to none." NASA's Deputy Administrator, Dr. Hugh L. Dryden, has spoken along the following lines: "The ultimate and practical purpose of these large expenditures is twofold: (1) insurance of the Nation against scientific and technological obsolescence in a time of explosive advances in science and technology; and (2) insurance against the hazard of military surprise in space."

Another commentator cites national pride as a justification of the space program. "It should be enough without all the other reasons and rationalizations that have been presented. It is the proper motivation of a prideful people with vitality, a sense of destiny, and confidence in their own ability."<sup>2</sup> Still another view is that we are engaged in "an effort with no practical, describable motive at all. An effort engaged in simply for its own sake. . . .

<sup>2</sup> Harold Leland Goodwin, *Space: Frontier Unlimited* (Princeton: Van Nostrand, 1962), p. 111.

Some will quarrel with the description of space exploration as a motiveless enterprise. And yet most of the motives advanced for it seem to me more like by-products than like major purposes."<sup>3</sup>

President Eisenhower never attempted a coherent statement of the rationale of the space program, and neither did President Kennedy. But some of Kennedy's statements might be cited, selected so as to suggest motives not brought out fully by the quotations already given. Once Kennedy referred to the congressional injunction that NASA should contribute to "the preservation of the role of the United States as a leader," and he said that it was "to meet that great responsibility that I have suggested a great national effort in the field of space for the American people." He thought that it was "time for this nation to take a clearly leading role in space achievement." According to him, the United States wishes "to demonstrate to a watching world that it is first in the field of technology and science. . . . I do not believe that we want to permit the Soviet Union to dominate space, with all that it might mean to our peace and security in the coming years." "If we can get to the moon before the Russians, we should."<sup>4</sup>

But why, some say, the moon? Why choose this as our goal? And they may well ask why climb the highest mountain. Why, thirty-five years ago, fly the Atlantic? Why does Rice play Texas?

We choose to go to the moon. We choose to go to the moon in this decade, and do the other things not because they are easy, but because they are hard, because that goal will serve to organize and measure the best of our energies and skills, because that challenge is one that we are willing to accept, one we are unwilling to postpone, and one we intend to win, and the others, too.

It is for these reasons that I regard the decision last year to shift our efforts in space from low to high gear as among the most important decisions that will be made during my incumbency in the office of the Presidency.<sup>5</sup>

A year later Kennedy's mood was quite different.

Why should man's first flight to the moon be a matter of national competition? Why should the United States and the Soviet Union, in preparing for such expeditions, become involved in immense duplications of research, construction, and expenditure? Surely we should explore

<sup>3</sup> Robert Colborn, "In Our Opinion," *International Science and Technology*, January, 1963, p. 19.

<sup>4</sup> U.S. President, *Public Papers of the Presidents of the United States*, John F. Kennedy . . . January 20 to December 31, 1961 (Washington, 1962), pp. 310, 403, 407, 734 (hereafter cited as *Public Papers of the Presidents*).

<sup>5</sup> *Public Papers of the Presidents*, Kennedy, 1962, p. 669.

whether the scientists and astronauts of our two countries—indeed of all the world—cannot work together in the conquest of space, sending someday in this decade to the moon, not the representatives of a single nation, but the representatives of all of our countries.

Now most of these statements of purpose can be fitted together; most of the motives cited may really be operating. Nevertheless there is enough variety and contradiction in the statements to arouse curiosity. What motives actually seem to operate? What differences are there in the motives endorsed by different persons and groups at different times? What unintended effects, if any, does the space program seem to be producing? Can a rational man find a rationale that permits him to support the program, or must he conclude that modifications of some sort are required? What does behavior relating to space tell us about our behavior—and about human behavior—in other realms?

## THE HISTORICAL DEVELOPMENT OF THE SPACE PROGRAM

# 2

The American record pertaining to space is both miserable and impressive. We might argue about the reasons for the miserable aspects of the record. To some extent there was a failure to foresee certain scientific and technological possibilities. Even after these possibilities became reasonably well established, there was a failure to see their potential importance in a psychological and political sense; and coupled with this was a certain blindness to evidence of growing Soviet capacities. Perhaps these reasons should suffice as an explanation, but they fall short in that they do not tell us the why of the failures and the blindness. Implicitly they raise the question whether there was something about the outlook of those mainly responsible that dulled sensitivities, whether there was a lack of a clear and coherent set of values that might have made them alert to relevant possibilities and developments. When people do not know what matters to them—what their goals are—failures and frustrations are not surprising.

There is less room for argument about the assertion that the record is impressive. A program of great magnitude is under way, flowing from legislation, from organizational arrangements, and from an awakened sense of purpose—all worth noting. Moreover, the program is producing striking results in various areas, and more are yet to come. Whether they are worth the costs, whether the resources being devoted to space might produce still more

and their work, and that in a democracy parties should pursue the voters; and it is inevitable that in a political system major decisions will be influenced by political considerations. The question is not whether special interests and ulterior motives are pursued but to what extent, if at all, they are the "real" motives, for which the public interests and manifest motives discussed in the preceding chapters are covers, and to what extent, if at all, they are pursued at the expense of these public interests and manifest motives. Given the magnitude of the space effort and the number of people involved, it is a foregone conclusion that for many the publicly expressed motives are rationalizations and that sometimes private advantage is pursued at the expense of the public good. But this kind of thing is surely the exception rather than the rule.

## THE RELATIVE IMPORTANCE OF DIFFERENT MOTIVES

# 11

The motives for developing the space program have varied in importance in the minds of different persons and groups and at different times. This general proposition is worth elaborating on, and a few more specific propositions can be made. I will speak on the basis of what is better described as an informed impression than as hard evidence.

From the very beginning motives were mixed. The urge to try it, to find out, to discover and explore, to do the new and the challenging lay back of the increasing interest in space in the decades and years leading up to Explorer I and Vanguard. Specifically scientific purposes were no doubt dominant in the minds of those planning the IGY. Those within the military structure concerned with the development of missiles (e.g., the Wernher von Braun group and some of those above it in the chain of command) shared these motives and also had some thoughts about the military potentialities of space and, perhaps, the implications of space exploits for national prestige. But prior to Sputnik the military and political leadership were not greatly impressed by these latter considerations, as is demonstrated by the meagerness of the funding provided for the Vanguard program.

For many of the natural scientists—especially those critical of the lunar program—the desire to enhance knowledge has remained the dominant reason for a space program; in truth, these scientists seem to find it difficult to grasp the fact that other reasons really count. For other natural scientists and for those outside the natural

sciences the scientific reasons for a space program have much less relative significance. Mr. Webb makes a formal distinction between scientific objectives, on the one hand, and the commitment to a lunar landing, on the other; and he clearly attaches much greater importance to the lunar landing. Of course, the distinction does not have as much practical significance as might be the case, for preparations for the lunar landing necessitate considerable scientific inquiry. Spokesmen for the military rarely stress the purely scientific purposes of the space program. When members of Congress do, they give the impression as often as not that they are rationalizing, e.g., that having decided to support a space program out of national pride, they justify it in the name of contributions to science. They would vote only a small fraction of current appropriations for a space effort if the enhancement of scientific knowledge were the only object.

The desire to enhance military security through a space program is influential in governmental circles—emphasized primarily by the military, by certain members of Congress whose general outlook calls for great stress on the armed forces and military preparedness, and to a much lesser extent by NASA officials. Concern for military security in relation to space seemed most intense in Congress in the first two or three years of the space age. To some extent this appears to have been a partisan matter, with Democrats looking very hard for Republican faults. To some extent it appears to have reflected confusion between space power and missile power. Soviet missiles have fundamentally transformed America's strategic position. How much the transformation is accentuated by actual or potential Soviet space power is as yet uncertain. Despite the fact of uncertainty, there have been many assertions that national survival is at stake in the space race, and fears for security (whether in the short or long run) have clearly had a very substantial influence in Congress. Gilbert Burck goes so far as to describe the whole space program as "para-military," but this is scarcely justified. True, the Air Force has an extensive space program, and there are potential military implications in most of what NASA does, even in the "purely scientific" investigations. But it was not the Air Force that induced Kennedy to go for the moon. If concern for military security were dominant, there would be no Project Apollo; there would instead be an emphasis on the development of various kinds of capabilities in terrestrial space. Thus, though the security

motive has been powerful, whether for good reasons or bad, it has been one among others, and probably not the most influential one. I will comment more on this below.

To some extent the emphasis on the military aspects of the space program has been restrained by a desire to stress peaceful purposes. In Chapter 5 I have noted the confusion of thought attending the word *peaceful*, concluding that the general intent seems to be to avoid creating alarm, to avoid the appearance of threatening anyone, to avoid contributing to a frame of mind in which thoughts of war are in the foreground. Peaceful purposes in this sense are emphasized mainly by political leaders in Congress and in the Administration, and not so much by others. Emphasis on the point has been fairly constant in time.

In political circles since Sputnik I concern for national prestige and pride has been the most powerful of the motives associated with the space program. And though prestige is more commonly named, more stress should go to pride. Signs of both motives, especially concern for prestige, show in the planning of the Vanguard program. Sputnik I brought them, along with the security motive, into the foreground, though they did not gain great strength as long as it was more or less complacently assumed that the United States would duplicate Soviet feats with some speed. But when successive Soviet firsts were matched by so much ignominy at Cape Canaveral, thoughts of lost prestige began to rankle, and the hurt to pride became galling. Eisenhower resisted the attitude. At the same time, he may inadvertently have strengthened it. He took a very restricted view of the role of the President, failing to provide exciting and dynamic leadership. Moreover, the forced cancellation of his trip to Japan, the blundering associated with the U-2 affair, and the indignities suffered by Vice President Nixon in Venezuela all raised distressing questions about both respect abroad and self-respect at home. Republicans as well as Democrats in Congress became increasingly disturbed, and when the Kennedy Administration came in, all were ready to join in an effort to recoup the situation, without much regard to cost.

The very act of adopting a bold and challenging program probably did something to relieve the psychological pressures that led to it. Further, even before Kennedy's inauguration, Cape Canaveral was producing a much greater measure of satisfaction than earlier in terms of successful launches and the acquisition of scientific

knowledge; and successes were also being scored from the Vandenberg Air Force Base in California. After the inauguration came the suborbital flights of Shepard and Grissom, and a year later the Glenn flight that did so much to purge the country of feelings of inferiority and doubt and to restore prestige and pride. Moreover, as if to compensate for the miseries of the Bay of Pigs came the confrontation with the Soviet Union over Cuba in October, 1962, and the tremendous surge of national confidence that success brought.

With national morale restored, however, concern for prestige and pride has lost some of its force as a motive for the space program. The change began to show in Congress in the spring of 1962, after the Glenn flight, and it has become increasingly apparent. Though many in Congress are no doubt still moved by a desire to beat the Russians, others have come to express scorn for this motive; and some favor making even the lunar program a cooperative enterprise. Thus a motive that once was so obsessive has declined significantly in its appeal. Whether or not Kennedy's proposal of September, 1963, reflected a genuine desire for cooperation with the Russians, it was logically inconsistent with the principal motives of the expanded program that he had championed two years earlier.

Speaking figuratively, the developments raise the question whether America knows its own mind. The question has been raised many times before. Time and again the country has demonstrated either an unawareness of values that turned out to matter very much or an inconstancy in pursuing them. We went into the Spanish-American War at the end of the last century without a consensus on the values being pursued and thus without a coherent conception of the strategy to employ and the terms of peace to insist upon when victory was won. In the years prior to World War I there was substantial unawareness of the importance to American values of averting such a conflict; and when the disaster engulfed Europe, Wilson declared our neutrality—only later to ask for a declaration of war in the name of goals that had nothing to do with those that he had been proclaiming up to that time. And there was no consensus on goals when victory came, with the result that we left it largely to others to determine what would or would not in the longer run be achieved. Then what were our goals between the wars and down to Pearl Harbor? Not knowing what we wanted, we got the appalling blight of Nazism and the tragedy of World War

II. For half a century each crisis that came upon the country found it without a clear conception of goals and how to achieve them, compelling improvisation that usually had dismal results.

The record in diplomatic affairs looks better since World War II, but previous failures warn against assuming too much. So far as space is concerned, there was an obvious general failure before Sputnik to identify values that soon come to seem so important. Even after Sputnik, many of those who were clearly motivated by pride gave other reasons for wanting to develop a major space program. Not until 1963 did it become more or less common for those analyzing the reasons for the space program to mention concern for pride or self-esteem among the major factors. And doubt about the justifiability of the program, particularly the projected manned lunar landing, became prominent at about the same time.

In part, the shift in attitudes may be due to increasing awareness of the influence of pride and to a genuine belief that the desire to bolster it through a lunar landing does not justify the great expenses involved. In part, the explanation is to be found in an analysis of human motivation by A. H. Maslow.<sup>1</sup> He speaks of motivation in terms of needs, suggesting a number of categories: physiological needs, safety needs, love needs, esteem needs, etc. People have motives in a number of such categories, but are rarely conscious of them all. When their physiological interests are threatened, concern for them is likely to be a controlling guide to behavior; for a man threatened with suffocation, a breath of air is the paramount motive. But when a reasonable degree of satisfaction for the physiological needs is assured, they are more or less taken for granted, and attention is directed to, e.g., safety needs. When satisfaction for these can also be taken for granted, they are more or less forgotten and attention shifts to another category. Maslow suggests that, in principle, there is an endless set of categories of motives or needs (goals, values, interests), and that men shift their attention back and forth among the categories depending on circumstances. Changes in circumstances may call attention to values that hitherto were ignored or may lead to the development of totally new values.

In other words, a value provides motivation mainly when it is in jeopardy. When there is no problem about it, no one gives it any

<sup>1</sup> A. H. Maslow, "A Dynamic Theory of Human Motivation," in Chalmers L. Stacey and Manfred F. DeMartino, eds., *Understanding Human Motivation* (Cleveland: Howard Allen, 1958), pp. 28 ff.

thought. Thus people may not become aware of a value until a threat to it develops; and if the threat recedes, concern for the value may abate. This is precisely what seems to have happened in connection with the space program. It took Sputnik to awaken the American people, and Congress, to some precious values—especially to the importance of pride; and with the restoration of pride, concern for it diminishes. Concomitantly, there is an increasing tendency to cite military security as a major motive, whether because of a genuine Soviet threat in space or because this kind of appeal usually elicits a favorable response.

**PART**  **METHODS**

*Though labeled "Motivations," Part II also dealt in part with methods, e.g., with the question of the extent to which space R&D and various uses of satellites are likely to be effective methods of promoting the various goals.*

*The chapters of Part III concern methods exclusively: how the federal government has organized itself for the implementation of the space program; the principles and the strategies that NASA employs in mobilizing the necessary resources; and international cooperation and the dissemination of information at home and abroad as methods of promoting security, peace, the enhancement of scientific knowledge, and the other goals.*

*The treatment of methods is incomplete. Scientific research and the development of hardware are important methods in the pursuit of space goals, but I must leave it to others to describe and appraise activities in these fields. And I do not propose to examine the recruitment and assignment of administrative personnel in NASA, or detailed management policies, even though what is done in these fields may have vital significance.*